

Mahad Rana

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EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Bachelor of Science in Computer Science

Expected: May 2026

- GPA: 4.0
- Honors: Dean's List, Edwin A. Stevens Scholarship, Presidential Scholarship, Pinnacle Scholars
- Relevant Courses: Data Structures, Algorithms, Discrete Structures, Deep Learning, Systems Programming

EXPERIENCE

Software Development Intern

May 2024 - August 2024

Croll-Reynolds

Parsippany-Troy Hills, NJ

- Increased application productivity by 25% by expanding a .NET program to handle 4-stage vacuum computations and automatically generate technical Excel datasheets.
- Boosted program security by implementing time-based locks and a serial number security layer, safeguarding sensitive data and preventing unauthorized access.
- Improved satisfaction by reducing the number of user-reported bugs by 30% through the resolution of issues in calculations for process vacuum system components within the application.

Undergraduate Machine Learning Researcher

May 2023 - August 2023

Stevens Institute of Technology

Hoboken, NJ

- Formulated image processing algorithms using the OpenCV library in Python to preprocess a dataset of 100+ high-resolution images, resulting in reduced noise and improved model performance.
- Implemented an image segmentation machine learning model using the PyTorch library in Python that can create binary contours from images of guidewires with an accuracy rate of 95%.
- Developed an image-processing algorithm using MATLAB that creates a smooth midline from a contour of a guidewire for further calculations.

PROJECTS

Full-Stack Prayer Tracker | *JavaScript/React, Node.js, Express.js, MySQL, HTML/CSS, Docker*

- Created a responsive frontend for a prayer tracking application using React, implementing advanced state management with Context API and custom hooks to provide a smooth user experience.
- Configured CRUD operations in RESTful API endpoints using Express.js to ensure seamless communication between the MySQL database and the user interface, providing accurate and timely updates.
- Utilized Docker to create isolated environments for the frontend, backend, and database services, streamlining development, deployment, and scalability.

English to Spanish Translator | *Python, TensorFlow*

- Accomplished effective machine translation as demonstrated by an average BLEU score of over 0.3 by developing a Seq2Seq model with a Bidirectional LSTM in Python using TensorFlow.
- Applied advanced text preprocessing techniques, including sequence conversion, tokenization, and one-hot encoding, to streamline the data pipeline and enhance the model's efficiency and accuracy.
- Conducted extensive hyperparameter tuning and model optimization, leveraging various evaluation metrics and validation techniques to achieve optimal translation performance.

HelpDesk AI | *AWS, JavaScript/React, Node.js, Express.js, HTML/CSS*

- Deployed and maintained a full-stack application on AWS EC2, ensuring secure communication between services by configuring security groups to control inbound and outbound traffic, and using PM2 for process management.
- Integrated GPT models with an Express.js backend using OpenAI's API, enabling intelligent responses for troubleshooting queries with a focus on secure and efficient request handling.
- Engineered a React frontend with real-time chat functionality and advanced state management, improving responsiveness and enabling efficient, interactive communication.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, HTML/CSS, JavaScript, MATLAB, R, Ocaml, Bash, VB.NET, XML, C#, SQL

Frameworks: React, Express.js, JUnit, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, Matplotlib, WPF, .NET

Tools: Git, AWS, MongoDB, Node.js, Linux, Ubuntu, Unix, Visual Studio, Docker